

An Elementary, Self-Certifying Avoidance Lemma: Short Lattice Vectors off Bounded-Degree Varieties by Union Counting over a Coefficient Box

Eric Schultz Claude (Anthropic)

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Abstract

We give a short, elementary proof that a lattice of covolume U contains a nonzero vector of controlled ℓ^∞ -norm avoiding the zero sets of finitely many polynomials of bounded degree, under the sole hypothesis that no polynomial vanishes identically on the lattice. The method is a union-bound count over an explicit integer box of basis coefficients: each degree- δ constraint kills at most $\delta(N+1)^{d-1}$ of the $(N+1)^d$ candidate coefficient tuples, so a box of side $N \gtrsim N_0\delta$ leaves a good tuple. The avoidance phenomenon itself is classical and effective versions are known through successive-minima machinery; our contribution is methodological. The argument needs only a single norm-bounded basis — no well-roundedness, no control of higher successive minima, no reduction theory beyond Minkowski’s second theorem — and every constant it produces is a finite quantity verifiable by direct enumeration. We show the linear ($\delta = 1$) case covers affine hyperplanes at no extra cost, that the through-origin hypothesis of the usual statement is never load-bearing, and that the degree factor δ is sharp: a product of δ parallel affine hyperplanes saturates the count exactly (a special case of the optimal degree dependence in Fukshansky [1]). All finite cores are confirmed by exhaustive enumeration. This note has not been externally refereed.

Provenance and priority

The results of this note are not new, and we make no claim to the contrary. The existence of a small-height point avoiding a union of varieties, with an explicit bound whose dependence on the height and on the degree of a containing hypersurface is *optimal*, is a theorem of Fukshansky [1], proved through Alon’s Combinatorial Nullstellensatz [2]. In particular, the sharpness of the degree factor recorded in our Proposition 1 is a special case of that optimality, not a new observation. Effective lattice versions avoiding full-rank sublattices together with an algebraic hypersurface appear in Fukshansky–Jeong [3], via Minkowski’s successive minima. The vector-space ancestor — a space over an infinite field is not a finite union of proper subspaces — is the classical finite-union lemma.

What this note offers is therefore not a theorem but a *packaging*: a bare-hands union-bound count over an explicit integer box of basis coefficients, implementing by hand the “a bounded-degree polynomial cannot vanish on a large enough grid” idea that the Combinatorial Nullstellensatz supplies in polished form. Its only virtues over the cited literature are pedagogical and computational: it requires just one norm-bounded basis rather than control of all successive minima or a Nullstellensatz invocation, it drops the well-roundedness and through-origin assumptions found in some treatments, and it produces only finite, enumerable constants, so every core inequality can be — and here is — confirmed by direct enumeration. The price is real: the constants are crude, a factor

of order d off the successive-minima route and with none of the optimality of [1]. This is a self-contained lemma to drop into an applied argument when a checkable existence certificate matters more than a sharp height, not a contribution to the avoidance literature.

1 Setup

Throughout, $L \subset \mathbb{R}^d$ is a lattice of rank $d \geq 2$ and covolume $U = \text{covol}(L) > 0$. For $x \in \mathbb{R}^d$ we write $\|x\|_\infty = \max_i |x_i|$.

Lemma 1 (Bounded basis). *Every rank- d lattice L of covolume U admits a basis $v_1, \dots, v_d$ with $\|v_i\|_\infty \leq C_1(d) U^{1/d}$ for all i , where $C_1(d)$ depends only on d .*

Proof. Minkowski's second theorem bounds the product of successive minima, $\lambda_1 \cdots \lambda_d \leq 2^d U / V_d$, with V_d the volume of the unit ℓ^∞ ball; classical reduction theory (Cassels [4], Ch. VIII) converts the successive minima into an actual basis bounded in terms of them. The crude bound $\|v_i\|_\infty \leq C_1(d) U^{1/d}$ suffices; we do not track the sharp constant. $\square$

We will use only the following consequence of the basis: every element of the sublattice generated by short combinations is short. Concretely, for a coefficient vector $c = (c_1, \dots, c_d) \in \mathbb{Z}^d$ with $\|c\|_\infty \leq N$, the vector $v(c) = \sum_i c_i v_i$ satisfies $\|v(c)\|_\infty \leq d N C_1(d) U^{1/d}$.

2 The avoidance lemma

We first record the polynomial input, then state and prove the lemma.

Lemma 2 (Grid Schwartz–Zippel). *Let $g \in \mathbb{R}[x_1, \dots, x_d]$ have total degree $\leq \delta$ and not vanish identically on $\mathbb{Z}^d$. Fix a nonzero linear map $c \mapsto v(c)$ from $\mathbb{Z}^d$ to $\mathbb{R}^d$ (given here by a lattice basis). If $g \circ v$ is not identically zero on $\{0, 1, \dots, N\}^d$, then $g \circ v$ vanishes on at most $\delta(N+1)^{d-1}$ of those $(N+1)^d$ tuples.*

Proof. $g \circ v$ is a polynomial in $c_1, \dots, c_d$ of total degree $\leq \delta$ (composition with the affine-linear v does not raise degree). The bound is the standard Schwartz–Zippel estimate over the finite grid $\{0, \dots, N\}$ per coordinate, valid over any integral domain: induct on d ; slice along c_d , where a nonzero univariate polynomial of degree $e \leq \delta$ has at most e roots in the grid, and apply the inductive bound with residual degree $\delta - e$ to the coefficient slices. See DeMillo–Lipton, Schwartz, and Zippel. $\square$

Lemma 3 (Avoidance). *Let L be a rank- d lattice of covolume U , $d \geq 2$, with a basis bounded as in Lemma 1. Let $g_1, \dots, g_N \in \mathbb{R}[x_1, \dots, x_d]$ have total degree $\leq \delta$, and suppose no g_j vanishes identically on L . Then there is a nonzero $v \in L$ with $g_j(v) \neq 0$ for every j and*

$$\|v\|_\infty \leq (N\delta + 1) d C_1(d) U^{1/d}.$$

Proof. Set $N' = \lceil N\delta + 1 \rceil$ and consider the coefficient box $\{0, 1, \dots, N'\}^d$, which has $(N' + 1)^d$ tuples, of which $(N' + 1)^d - 1$ are nonzero. Fix j . Because g_j does not vanish identically on L , the composition $g_j \circ v$ is not identically zero on $\mathbb{Z}^d$; in particular we may assume it is not identically zero on the box (if it were, enlarge N' by the finite amount needed to witness a non-root, which exists since $g_j \circ v \neq 0$). By Lemma 2, $g_j \circ v$ vanishes on at most $\delta(N' + 1)^{d-1}$ tuples. By the union bound, the number of tuples killed by *some* g_j is at most $N\delta(N' + 1)^{d-1}$. Since N and δ are

integers, $N' = N\delta + 1$ exactly, so $N' + 1 - N\delta = 2$; hence the number of surviving nonzero tuples is *exactly*

$$[(N' + 1)^d - 1] - N\delta(N' + 1)^{d-1} = (N' + 1)^{d-1}[(N' + 1) - N\delta] - 1 = 2(N' + 1)^{d-1} - 1,$$

which for $d \geq 2$, $N, \delta \geq 1$ (whence $N' + 1 \geq 3$) is at least $2 \cdot 3^{d-1} - 1 \geq 5 > 0$. Any surviving tuple $c^* \neq 0$ yields $v = v(c^*) \neq 0$ with $g_j(v) \neq 0$ for all j and, by the coefficient bound following Lemma 1, $\|v\|_\infty \leq d N' C_1(d) U^{1/d} \leq (N\delta + 1) d C_1(d) U^{1/d}$. $\square$

Remark 1 (Slack in the count). The proof produces not one good tuple but at least $2(N' + 1)^{d-1} - 1$, since the bracket $(N' + 1) - N\delta$ equals 2 identically rather than the ≥ 1 that existence requires. This is a genuine factor-of-2 margin in the union-bound count, and the true margin is larger still: when the forbidden zero sets share parallel components across different g_j , their bad tuples overlap and the union bound overcounts the damage, so the number of survivors strictly exceeds $2(N' + 1)^{d-1} - 1$ in those cases. Direct enumeration confirms both the exact count in the disjoint case and the surplus under overlap (Section 4). None of this changes the conclusion — a single good tuple already suffices — but it shows the box side $N' = N\delta + 1$ is chosen with room to spare, and could be shrunk if a sharper height were the goal.

3 Specializations and sharpness

Corollary 1 (Affine hyperplanes are free). *Let $H_1, \dots, H_N$ be affine hyperplanes $H_j = \{x : \ell_j(x) = a_j\}$ with each linear functional $\ell_j \neq 0$ not vanishing identically on L . Then there is a nonzero $v \in L$ off every H_j with $\|v\|_\infty \leq (N + 1) d C_1(d) U^{1/d}$.*

Proof. Apply Lemma 3 with $\delta = 1$ and $g_j(x) = \ell_j(x) - a_j$. Degree 1 gives $N' = N + 1$ and the stated bound. $\square$

Remark 2 (The through-origin hypothesis is never load-bearing). The classical statement takes hyperplanes through the origin ($a_j = 0$). The proof of Lemma 3 never uses $a_j = 0$: the count of bad tuples for a single constraint depends only on the degree of $g_j \circ v$, not on its constant term. A nonconstant affine function of the coefficients hits any prescribed value at most once along each coordinate line, whether that value is 0 or not. The origin case is in fact very slightly *worse* for counting, not better: when H_j passes through the origin, the tuple $c = 0$ is one of its roots, so among nonzero tuples a through-origin hyperplane carries at most $(N + 1)^{d-1} - 1$ bad points, whereas a generic affine hyperplane can carry the full $(N + 1)^{d-1}$. This costs nothing in the conclusion.

Remark 3 (Lower-dimensional targets are easier). To force $v \notin V$ for a linear subspace V of codimension ≥ 1 , it suffices to force v off any single hyperplane $H \supseteq V$; avoiding a smaller set is never harder. Thus Corollary 1 subsumes avoidance of arbitrary positive-codimension affine subspaces at the same bound.

Proposition 1 (The degree factor is sharp). *For every $\delta \geq 1$ there is a degree- δ polynomial g , not identically zero on L , whose zero set meets the box $\{0, \dots, N\}^d$ in exactly $\delta(N + 1)^{d-1}$ tuples for all $N \geq \delta$. Hence the per-constraint bound of Lemma 2, and with it the degree factor in Lemma 3, cannot be improved in general.*

Proof. Take $g(x) = \prod_{r=1}^{\delta} (\ell(x) - r)$ for a coordinate functional $\ell(v(c)) = c_1$ (available when the basis is the standard one; in general use any functional that is a nonzero integer combination of coordinates of $v(c)$). Its zero set is the union of the δ parallel affine hyperplanes $c_1 \in \{1, \dots, \delta\}$.

For each of the δ admissible values of c_1 and each of the $(N+1)^{d-1}$ choices of the remaining coordinates, exactly one tuple lies on the zero set, and these are distinct, giving exactly $\delta(N+1)^{d-1}$ bad tuples. $\square$

3.1 A congruence specialization: CRT-prescribed residue patterns

The avoidance lemma is stated over $\mathbb{R}$, but its only inputs are a lattice, a bounded basis, and constraints that are polynomial in the basis coefficients. Congruence constraints fit this template, and combining Lemma 3 with the Chinese Remainder Theorem yields a short lattice vector realizing a prescribed residue pattern across several coprime moduli. We record this because it is the setting from which the underlying construction originally arose, and because the correct hypothesis over *prime-power* moduli is a genuine sharpening of the naïve one.

Corollary 2 (CRT residue realizability). *Let $U = \prod_{i=1}^r p_i^{e_i}$ with distinct primes p_i and known factorization. Let L be a rank- d lattice with a bounded basis as in Lemma 1, and for each i let ℓ_i be an integer-linear coordinate functional that is primitive modulo p_i , meaning its coefficient vector is not entirely divisible by p_i . For each i prescribe either a target residue $a_i \in \mathbb{Z}/p_i^{e_i}\mathbb{Z}$ or a forbidden set $F_i \subsetneq \mathbb{Z}/p_i^{e_i}\mathbb{Z}$. Then there is a nonzero $v \in L$ realizing every local constraint simultaneously — $\ell_i(v) \equiv a_i$, respectively $\ell_i(v) \notin F_i \pmod{p_i^{e_i}}$ — with $\|v\|_\infty \leq C(d, r) U^{1/d}$.*

Proof. Primitivity modulo p_i makes ℓ_i surject onto $\mathbb{Z}/p_i^{e_i}\mathbb{Z}$: a coefficient that is a unit modulo p_i is a unit modulo $p_i^{e_i}$, so ℓ_i already attains every residue as its argument ranges over a complete set modulo $p_i^{e_i}$. Hence each local constraint is satisfiable — the target a_i is attained, and any $F_i \subsetneq \mathbb{Z}/p_i^{e_i}\mathbb{Z}$ leaves a nonempty admissible complement. Since the $p_i^{e_i}$ are pairwise coprime, the Chinese Remainder Theorem assembles the local admissible classes into a single admissible union of residue classes modulo U ; realizing it is a system of congruences on v , i.e. the condition that v lie in a prescribed union of cosets of the congruence sublattice $L_U = \{v \in L : \ell_i(v) \equiv 0 \pmod{p_i^{e_i}} \forall i\}$. When the ℓ_i read independent coordinates, $[L : L_U] = \prod_i p_i^{e_i} = U$, so $\text{covol}(L_U) = U \text{ covol}(L)$. Each forbidden or off-target residue is an affine-congruence condition, hence a $\delta = 1$ constraint in the sense of Corollary 1 once lifted to the integer basis of L_U ; applying Lemma 3 over L_U produces a nonzero v meeting all of them with $\|v\|_\infty \leq C(d, r) U^{1/d}$. $\square$

Remark 4 (Primitivity is the right hypothesis, and it is sharp). Over a prime modulus $\mathbb{Z}/p\mathbb{Z}$ — a field — “nonzero functional” suffices, exactly as in Corollary 1. Over a *prime power* $\mathbb{Z}/p^e\mathbb{Z}$ with $e \geq 2$ this fails: the ring has zero divisors, and a functional that is nonzero modulo p^e but has all coefficients divisible by p is *not* surjective. Its image is the subgroup of multiples of p , of size p^{e-1} rather than p^e , so residues outside $p\mathbb{Z}/p^e\mathbb{Z}$ are unreachable. Direct enumeration confirms the collapse: $2x_0 \bmod 4$ has image $\{0, 2\}$ (index 2, not 4), and $3x_0 \bmod 9$ has image $\{0, 3, 6\}$ (index 3, not 9); whereas the primitive $3x_0 + 2x_1 \bmod 8$, with 3 a unit, surjects (index 8). The correct hypothesis is therefore primitivity modulo p_i , not mere nonvanishing modulo $p_i^{e_i}$ — the prime-power analogue of the “ $\ell_j \neq 0$ on L ” condition of Lemma 3. This is also what keeps the specialization honest about its scope: the reassembly across moduli is the Chinese Remainder Theorem, used as stated; the content contributed here is the height bound on the realizing vector together with the sharp local hypothesis. The general composite-modulus problem without a known factorization is harder and, for higher degree, is only settled in the multilinear regime (Bünz–Fisch); here the factorization of U is given, which is what makes the clean local-to-global passage available.

4 Verification

Every finite core above was checked by exhaustive enumeration over the coefficient box with the standard basis $v_i = e_i$ in dimension $d = 3$; the counts reported are exact.

Affine, $\delta = 1$ ($d = 3$, $N = 2$; box has 26 nonzero tuples). The through-origin hyperplane $\{c_1 = 0\}$ carries 8 bad nonzero tuples, matching $(N + 1)^{d-1} - 1 = 8$; its affine shift $\{c_1 = 1\}$ carries $9 = (N + 1)^{d-1}$. The adversarial functional $\{c_3 = 0\}$, which vanishes on two of the three basis vectors, also carries exactly 8, confirming that a constraint vanishing on part of the basis is still controlled through the remaining coordinate. A generic affine plane $\{c_1 + c_2 + c_3 = 3\}$ carries 7. All are $\leq (N + 1)^{d-1} = 9$.

Degree $\delta = 2$ ($d = 3$, $N = 2$; per-constraint budget $\delta(N + 1)^{d-1} = 18$). Irreducible conics are far from tight ($c_1^2 - c_2$: 5 bad; $c_1 c_2 - 1$: 3; $c_1^2 + c_2^2 - c_3$: 3). The reducible extremal cases saturate: $c_3(c_3 - 1)$ (two parallel planes) gives 17, and $(c_1 - 1)(c_1 - 2)$ gives exactly 18, meeting the bound with equality and confirming Proposition 1.

Sharpness across scale. For the extremal $g = (c_1 - 1)(c_1 - 2)$ with $d = 3$, direct enumeration gives bad-tuple counts 18, 32, 50, 72, 162 at $N = 2, 3, 4, 5, 8$, each equal to $2(N + 1)^2$ on the nose, while the surviving-tuple count $(N + 1)^3 - 1 - 2(N + 1)^2$ equals 8, 31, 74, 143, 566 respectively — strictly positive throughout, as Lemma 3 guarantees.

Survivor count and slack (Remark 1). Under the lemma’s own construction $N' = N\delta + 1$, the number of good nonzero tuples was enumerated directly. In the disjoint case it equals $2(N' + 1)^{d-1} - 1$ exactly: for $(N, \delta, d) = (1, 1, 2)$ this is 5; for $(1, 1, 3)$, 17; for $(2, 1, 3)$ and $(1, 2, 3)$, 31; for $(2, 2, 3)$, 71. When forbidden components overlap across constraints the survivor count is strictly larger — e.g. $(2, 1, 3)$ rises from the formula’s 31 to 35, and $(2, 2, 3)$ from 71 to 95 under δ parallel planes per constraint — confirming the union bound is conservative. In every case the minimum over admissible parameters is 5, never the bare 1 that existence needs.

Congruence specialization (Corollary 2, Remark 4). The index of the congruence sublattice and the primitivity trap were checked by direct enumeration. For independent coordinates with coprime moduli $(4, 9)$, the congruence sublattice of $\mathbb{Z}^2$ has index $36 = 4 \cdot 9$, the full CRT value. The non-primitive functionals collapse as predicted: $2x_0 \bmod 4$ has image size 2 and $3x_0 \bmod 9$ has image size 3, so neither surjects, while the primitive $3x_0 + 2x_1 \bmod 8$ has image size 8 and does. These confirm both the covolume identity $\text{covol}(L_U) = U \text{covol}(L)$ used in the corollary and the necessity of the primitivity hypothesis.

5 Discussion

The value of this note is not a new bound but a proof one can hand to a non-specialist and a machine at the same time. The successive-minima route gives sharper constants; the box-counting route gives a statement whose every quantity is a finite integer count and whose hypotheses reduce to “the constraints are not identically satisfied.” In settings where one needs a short lattice vector in general position — off a singular locus, off a low-rank configuration, off a prescribed union of forbidden patterns — and cares more about a checkable existence certificate than about optimal height, this elementary packaging may be the more convenient one to reproduce inline. The affine-is-free and degree factor observations delimit what the box count costs: nothing for affine shifts, and a factor of δ for degree — the latter being, as [1] already establishes, exactly the optimal dependence and no artifact of the crude method.

References

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